Recap

DNS services are affected by security risks involving the infrastructure and the database

It is important to ensure the integrity and authenticity of the mappings of names to values provided in the DNS response messages

DNSSEC is a security extension that provides validation mechanisms

DNSSEC Chain of Trust

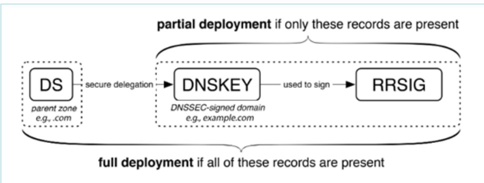
The chain of trust is the fundamental mechanism that lets DNSSEC-aware resolvers verify the authenticity and integrity of DNS data

It is a cryptographic pathway that follows the DNS hierarchical structure

The pathway consists of a series of digital signatures that link each level of the DNS system, from root to leaf, and allow the verification of the authenticity of the DNS data

The implementation is based on the `DS` (Delegation Signer) record type which contains a hash of the `DNSKEY` of a domain and is uploaded to the parent domain

Partial deployments cause problems



DNSSEC adoption

DNSSEC adoption is not very extensive (most TLDs, but only few second level domains are DNSSEC-enabled)

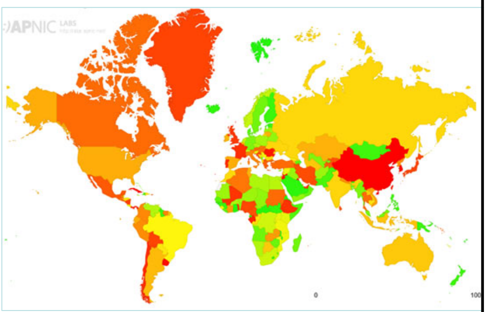
Validation rate is 34.4% worldwide and 27.2% for Italy

Many technical issues related to DNSSEC deployment

Complexity in the configuration and

Computational load and increased response time (more queries)

Potential vulnerabilities and security threats



[Source: https://stats.labs.apnic.net/dnssec]

DNS Privacy with Speed

DNS traffic is still largely unencrypted (DNSSEC does not ensure confidentiality), thus allowing cybercriminals (and ISPs) to derive user profiles from plaintext DNS queries

The solutions proposed to ensure privacy in the lookup phases are based on the encryption of query and response messages

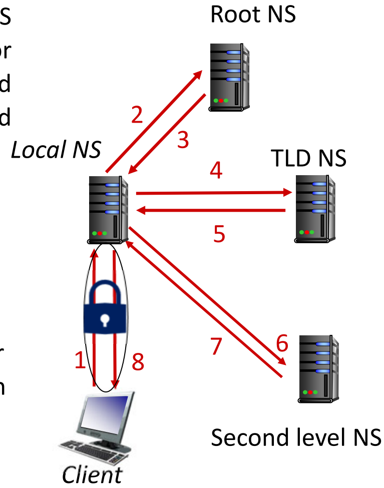
- DNS over TLS (DoT)
- DNS over HTTPS (DoH)
- DNS over QUIC (DoQ)

DoT and DoH are based on TCP transport protocol and on Transport Layer Security (TLS) encryption protocol

DoQ is based on the new QUIC transport protocol that combines transport and cryptographic features and ensures minimal latency

DNS over TLS (DoT)

The idea is to implement DNS lookups over TLS to ensure privacy [RFC 7858 “Specification for DNS over Transport Layer Security (TLS)” and RFC 8310 “Usage Profiles for DNS over TLS and DNS over DTLS” published in 2016 and 2018]

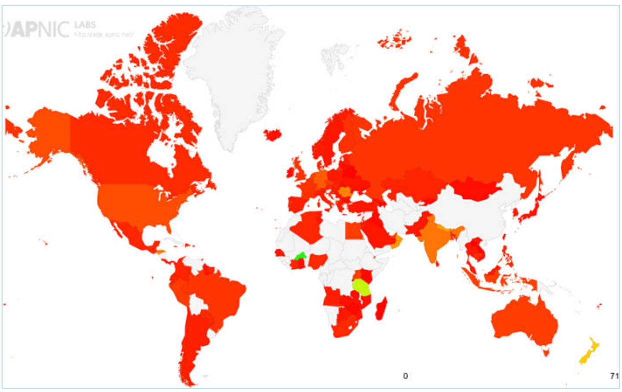


It complements DNSSEC and eliminates opportunities of eavesdropping and tampering of DNSSEC validated results on the “first leg”

Stub and recursive resolvers communicate over TLS-encrypted (permanent) TCP connections on port 853, thus ensuring channel integrity

Stub resolvers can be configured with strict privacy profiles or opportunistic privacy profiles to support incremental deployment

DoT adoption



[https://stats.labs.apnic.net/edns?s=2]

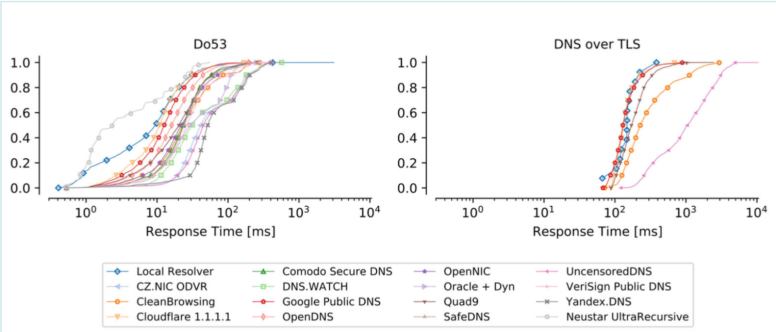
Only about 9.2% of the DNS queries are based on DoT

Some public recursive resolvers support DoT (e.g., **`cloudflare-dns.com`**, **`dns.google`**, **`dns.quad9.net`**, **`unfiltered.joindns4.eu`**)

DoT Performance

Measurements collected on the RIPE Atlas probes

Queries performed using DoT are slower by more than 100 ms compared to standard DNS queries (Do53) (e.g., overhead for the TCP and TLS handshakes)



[Source RIPE Atlas]

DNS over **`https`** (DoH)

Existing firewall configurations could block DNS queries over TLS

The idea is to implement DNS within **`https`** to ensure privacy and confidentiality [RFC 8484 “DNS Queries over HTTPS (DoH)” published in Oct. 2018]

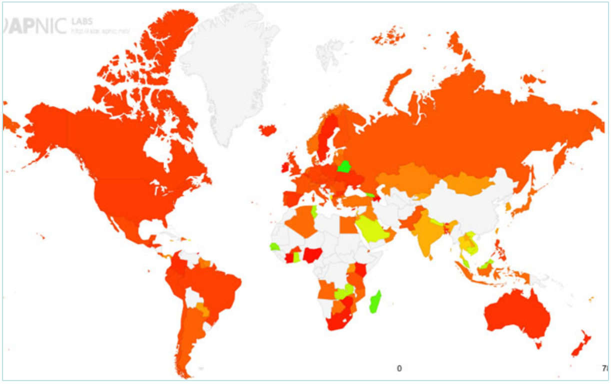
Each DNS query/response pair is mapped into an HTTP request/response pair

Queries are implemented using a **`GET`** method (i.e., content encoded in the URL) or a **`POST`** method (i.e., content included in the body as MIME type specified in **`Content-type`** header)

Standard port 443

Responses are sent as HTTP response messages

DoH adoption



[https://stats.labs.apnic.net/edns?s=1]

Only about 17.4% of the DNS queries are based on DoH

Most browsers support DoH with pre-defined public resolvers (e.g., Google, Cloudflare, NextDNS, DNS4EU)

Examples of query/response

Query: **pel.unipv.it IN A**

```
:method = POST
:scheme = https
:authority = dnsserver.example.net
:path = /dns-query
accept = application/dns-message
content-type = application/dns-
message
content-length = 33

pel.unipv.it IN A
```

Response: **193.204.34.1**
TTL= **3709** seconds

```
:status = 200
content-type = application/dns-
message
content-length = 61
cache-control = max-age=3709

193.204.34.1
```

DoH advantages & disadvantages

DoH provides very good privacy because unlike DoT it hides the service being accessed

DNS servers must be operated by trusted partners (they will see all queries)

Google, Cloudflare and Mozilla are pushing this protocol to promote privacy

DoH creates problems to ISPs and governments

Web surveillance (e.g., parental control, malware/website blocking) is often implemented by looking at DNS queries

In many countries ISPs are requested by law to store (for at least a year) the websites visited by citizens (e.g., monitoring of terrorism or other illegal activities)

Delivering content through CDNs becomes rather challenging

DNS over QUIC (DoQ)

DNS over QUIC [RFC 9250 “DNS over Dedicated QUIC Connections” published in May 2022]

Quick UDP Internet Connections (QUIC) is a UDP-based, stream-multiplexing, encrypted transport protocol

The encryption provided by QUIC has similar properties to those provided by TLS

DoQ has privacy properties similar to DoT and latency characteristics similar to classic DNS over UDP

DoT vs DoH vs DoQ

All these new protocols add desirable privacy and security features to DNS resolution

DoT is supported by operating systems (e.g., Linux, Android, MS Windows) and can be easily enabled, but its adoption rate is still quite low

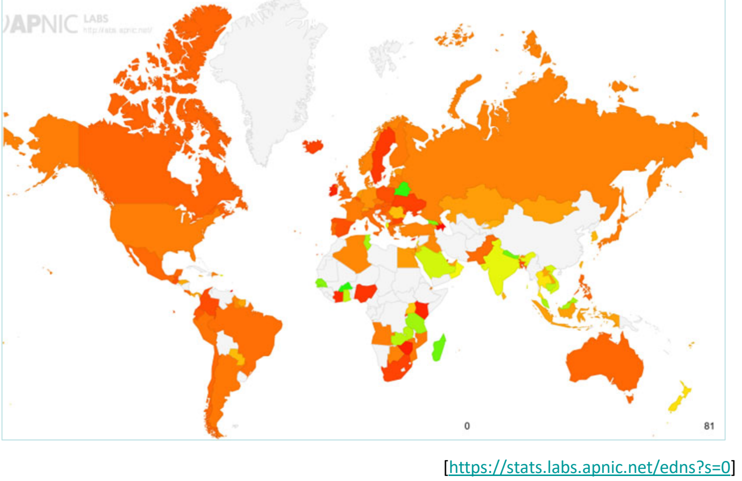
DoH can be easily enabled inside browsers

The impact of DoT or DoH on applications is generally not negligible

Geographic distance of recursive resolvers strongly affects the latency

In general DoQ outperforms DoT and DoH in terms of DNS single query response time and its adoption by public DNS resolvers is slowly increasing

Encrypted DNS



[https://stats.labs.apnic.net/edns?s=0]

About 26.6% of the worldwide DNS queries are encrypted

Europe is at 19.2%, while Asia is at 34.7%